

Nishant Chandraker

+91-9757064016 | nishant31.chandraker@gmail.com | [LinkedIn](#) | [GitHub](#) | [Leetcode](#)

SUMMARY

Software Engineer with experience in backend development, database optimization, and scalable system design. Proficient in building RESTful APIs, optimizing data-intensive applications, and developing full-stack systems. Strong foundation in data structures, algorithms, and distributed systems with hands-on experience in real-world and system-driven projects.

EDUCATION

Vellore Institute of Technology

B.Tech - Computer Science and Engineering, 7.44/10 CGPA

Vellore, TN, IND

July 2020 - June 2024

EXPERIENCE

Tata Consultancy Service - System Engineer

Aug 2024-Dec 2025

Finance

- Developed and enhanced **backend data** processing modules using **PL/SQL and .NET**, supporting financial systems handling **100K+ records per day**.
- Optimized **complex SQL queries, stored procedures, and batch workflows**, reducing execution time by **25-35%**.
- Contributed to **end-to-end feature enhancements** across **3+ data products**, from requirement analysis to deployment.
- Built reusable validation and transformation logic, reducing **data inconsistencies by ~30%**.
- Collaborated with **5+ cross-functional stakeholders** (business, QA, frontend) to deliver scalable solutions.
- Resolved **15+ production issues**, performing root cause analysis and implementing long-term fixes.
- Improved system observability by introducing logging and validation checks, reducing debugging time by **20%**.

Bhabha Atomic Research Center - Software Developer Intern

May 2023 - July 2023

Web service

- Designed and implemented a **normalized relational database schema** to manage structured employee data efficiently.
- Developed **RESTful backend services using Spring Boot**, enabling seamless data exchange between frontend and backend systems.
- Implemented business logic for data validation, retrieval, and transformation across multiple modules.
- Optimized database queries and indexing strategies to improve **data retrieval performance**.
- Participated in designing modular service architecture for better scalability and maintainability.

PROJECTS

Cloud-Native File Storage & Sharing System (GCP) |

GCP (Cloud Run, Cloud Storage, Cloud Functions), Node.js, Firestore, CI/CD

- Built a scalable file storage system supporting **10K+ uploads** using Cloud Storage.
- Deployed backend APIs on Cloud Run with automatic scaling.
- Implemented serverless workflows using Cloud Functions.
- Designed **CI/CD pipeline using GitHub Actions** for automated testing and deployment.
- Improved performance and security using caching and access control mechanisms.

Real-Time Smart Traffic & Route Optimization System |

Java, Node.js, PostgreSQL, Redis, Apache Kafka, REST APIs

- Designed a **real-time traffic monitoring and route optimization system** to suggest fastest routes based on live traffic conditions.
- Built backend services to process and analyze simulated **GPS data streams (100K+ events/day)** using **Apache Kafka**.

- Implemented **graph-based shortest path algorithms (Dijkstra/A*)** to dynamically compute optimal routes.
- Developed **REST APIs** to deliver route recommendations with **low latency (<150ms)**.
- Integrated **Redis** caching to store frequently accessed routes, reducing response time by **40%**.
- Designed normalized database schema in PostgreSQL for storing road networks, traffic data, and user queries.
- Simulated real-world traffic congestion scenarios to test system scalability and fault tolerance.

Distributed Chat Application (Real-Time Messaging System) |

Node.js, WebSockets, MongoDB, Redis

- Developed a real-time messaging system supporting **instant communication and message delivery guarantees**.
- Implemented **WebSocket-based communication** for low-latency messaging.
- Designed message queue and retry mechanisms to ensure reliable message delivery.
- Used Redis for session management and caching active user connections.

TECHNICAL SKILLS AND COMPETENCIES

Languages: Java, Python, SQL, JavaScript
Backend: Spring Boot, FastAPI, Node.js, .NET
Databases: Oracle, PostgreSQL, MySQL, Firestore
Cloud & DevOps: GCP, Docker, Terraform, CI/CD, GitHub Actions, Airflow
Core CS: Data Structures & Algorithms, System Design, Distributed Systems, Database Design, REST APIs, Performance Optimization